Olympiad Test Paper — Mathematics (Class 7)

Chapter 8: Working with Fractions

Subject	Mathematics (NCERT Class 7)
Chapter	8 — Working with Fractions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember that "of" means multiply, and to divide by a fraction you flip the DIVISOR and multiply. Watch the classic traps: multiplying a whole number into a fraction does NOT change the denominator (3 × 1/4 = 3/4, not 3/12); multiplying by a fraction below 1 makes things smaller while dividing by it makes them bigger; and mixed numbers must be turned into improper fractions before you multiply or divide. Always reduce your final answer. Do not write on this paper — mark answers on your answer sheet.

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1. The phrase "the product of 1/2 and 1/3" equals:

(A) 5/6 (B) 1/6
(C) 2/3 (D) 1/5

2. 3 × 1/4 equals:

(A) 3/12 (B) 1/12
(C) 4/3 (D) 3/4

3. Which expression correctly represents "2/5 of 30"?

(A) 2/5 ÷ 30 (B) 2/5 + 30
(C) 2/5 × 30 (D) 30 ÷ 2/5

4. The reciprocal of 3/7 is:

(A) 7/3 (B) −3/7
(C) 3/7 (D) 1/3

5. Compute 7 × 3/5 and write it as a mixed number.

(A) 21/35 (B) 10/5
(C) 4 1/5 (D) 3 2/5

6. What is 2/3 of 3/5?

(A) 5/8 (B) 6/8
(C) 2/5 (D) 2/15

7. Compute 5/6 ÷ 2/3.

(A) 5/9 (B) 10/18
(C) 4/5 (D) 5/4

8. How many 1/2-cup servings can be poured from 3 cups of juice?

(A) 6 (B) 3/2
(C) 1 1/2 (D) 3

9. Which of these multiplications gives a result SMALLER than 8?

(A) 8 × 5/4 (B) 8 × 1
(C) 8 × 3/4 (D) 8 × 9/8

10. The reciprocal of the whole number 5 is:

(A) 5 (B) 1/5
(C) −5 (D) 0

11. Dividing a number by which of the following is the same as multiplying it by 4/5?

(A) 4/5 (B) 5
(C) 5/4 (D) 20

12. A rectangle has length 2/3 m and breadth 3/4 m. Its area (in m^2) is:

(A) 1/2 (B) 5/7
(C) 17/12 (D) 6/7

13. Convert the mixed number 2 1/2 to an improper fraction.

(A) 4/2 (B) 5/2
(C) 3/2 (D) 21/2

14. Compute 2 1/2 × 1 1/3 by first converting to improper fractions.

(A) 2 1/6 (B) 3 1/3
(C) 2 1/2 (D) 3 2/3

15. A 4 1/2-acre plot is divided into 3/4-acre shares. How many shares are there?

(A) 3 (B) 12
(C) 6 (D) 4 1/2

16. Which statement is TRUE?

(A) Multiplying any number by 2/3 makes it larger.
(B) Dividing 6 by 1/2 gives 3. (C) Dividing a number by 1/4 multiplies it by 4.
(D) The reciprocal of 1 is 0.

17. Half OF a third equals:

(A) 5/6 (B) 1/6
(C) 2/3 (D) 1/3

18. Ahaan spends 1/3 of his money on a book, then 1/4 of what remains on a pen. What fraction of his original money is left?

(A) 5/12 (B) 1/4
(C) 2/3 (D) 1/2

19. A recipe needs 1 1/2 cups of flour per cake. How many cups are needed for 2 2/3 cakes?

(A) 4 (B) 4 1/2
(C) 3 (D) 4 1/6

20. Compute $3/4 \times 20$.

(A) 15 (B) 80
(C) 60 (D) 5

21. Which product is the LARGEST?

(A) $12 \times 1/2$ (B) $12 \times 2/3$
(C) $12 \times 3/4$ (D) $12 \times 5/6$

22. The value of $4/9 \times 3/8$, in lowest terms, is:

(A) 12/72 (B) 1/6
(C) 7/17 (D) 1/12

23. A tortoise walks 1/4 km each hour. How far does it walk in 3 1/2 hours?

(A) 3 1/2 km (B) 14 km
(C) 1 1/8 km (D) 7/8 km

24. $1 \div 2/3$ equals:

(A) 2/3 (B) 3/2
(C) 1/3 (D) 2

25. Two-thirds of the students in a class of 30 wear glasses, and 1/5 of those who wear glasses also wear caps. How many students wear both?

(A) 20 (B) 6
(C) 4 (D) 5

26. Which of the following equals $2/3 \div 4$?

(A) 8/3 (B) 2/12
(C) 2/7 (D) 1/6

27. A pole 6 m long is painted; 2/3 of it is red and 1/2 of the rest is blue. What length (in m) is blue?

(A) 4 (B) 3
(C) 2 (D) 1

28. Which pair of fractions are reciprocals of each other?

(A) 2/5 and 5/2 (B) 3/4 and 3/4
(C) 2/3 and 3/4 (D) 1/2 and 2/4

29. A 3/4 m ribbon is cut into pieces each 1/8 m long. How many pieces are obtained?

(A) 3/32 (B) 6
(C) 8 (D) 4

30. Without computing exactly, which is true about 15 × 4/5?

(A) It is more than 15. (B) It equals 15.
(C) It is less than 15. (D) It is less than 4/5.

31. Compute (2/3) × (9/10) × (5/6).

(A) 1/2 (B) 16/19
(C) 90/180 (D) 3/5

32. A jug holds 2 1/4 litres. How many 3/8-litre glasses can be filled from it?

(A) 6 (B) 8
(C) 5 (D) 27/32

33. Three-quarters of a number is 18. The number is:

(A) 13 1/2 (B) 72
(C) 27 (D) 24

34. The expression "5 ÷ 1/3" gives the same result as:

(A) 5 × 1/3 (B) 5 + 3
(C) 5 × 3 (D) 1/3 × 5

35. Which of these is equal to 1?

(A) 7/8 × 8/7 (B) 3/4 + 1/4 × 0
(C) 2/3 ÷ 2/3 × 1/2 (D) 5/6 × 6/5 × 1/2

36. 1/2 of 1/3 of 1/4 equals:

(A) 3/9 (B) 3/24
(C) 1/9 (D) 1/24

37. A car covers 3/5 of a 200 km journey before lunch. How many kilometres remain after lunch?

(A) 120 (B) 40
(C) 60 (D) 80

38. Compute 2 1/3 ÷ 1 1/6.

(A) 14/18 (B) 2
(C) 49/18 (D) 1 1/2

39. Which value makes the sentence true: 3/8 × ____ = 3/8?

(A) 0 (B) 8/3
(C) 1 (D) 3/8

40. A wall is painted at 2/5 of a wall per hour. How long, in hours, to paint the whole wall?

(A) 2/5 (B) 2
(C) 7/5 (D) 5/2

41. Of the cake, Riya ate 1/4 and Sam ate 2/3 of what was left. What fraction of the whole cake did Sam eat?

(A) 1/2 (B) 11/12
(C) 2/3 (D) 1/4

42. The product 5/12 × 8 equals:

(A) 5/96 (B) 40/96
(C) 13/12 (D) 10/3

43. Which statement about reciprocals is FALSE?

(A) The reciprocal of a proper fraction is greater than 1.
(B) Zero has no reciprocal. (C) The reciprocal of 1 is 1.
(D) Every fraction is larger than its reciprocal.

44. A tank fills 3/4 of its volume in 1 hour. At the same rate, what fraction fills in 1/3 hour?

(A) 9/4 (B) 3/7
(C) 13/12 (D) 1/4

45. Solve: (1/2 + 1/3) × 6/5.

(A) 1 (B) 5/6
(C) 6/5 (D) 11/30

46. A garden is 5/6 of an acre; 3/10 of it is planted with roses. The rose area (in acres) is:

(A) 8/16 (B) 4/5
(C) 1/4 (D) 15/16

47. If 2/3 of a number equals 1/2 of another number, and the first number is 9, the second number is:

(A) 6 (B) 12
(C) 3 (D) 27/4

48. A piece of cloth 7/8 m long is shared equally among 7 children. Each child gets (in m):

(A) 7/8 (B) 1/8
(C) 49/8 (D) 1/7

49. Which division gives the LARGEST result?

(A) 6 ÷ 1/2 (B) 6 ÷ 2
(C) 6 ÷ 3 (D) 6 ÷ 1

50. Compute 1 1/4 × 4/5.

(A) 1 (B) 5/4
(C) 4/5 (D) 25/16

51. A worker paints 2/7 of a fence in a day. What fraction is left unpainted after 2 days?

(A) 4/7 (B) 2/7
(C) 3/7 (D) 5/7

52. The area of a square of side 2/3 cm is:

(A) 8/3 cm^2 (B) 4/6 cm^2
(C) 4/9 cm^2 (D) 4/3 cm^2

53. By what number must 3/5 be multiplied to give 1?

(A) 5/3 (B) 3/5
(C) 5 (D) 3

54. A bottle holds 3/4 litre. Two-thirds of it is poured out. How much (in litre) is poured out?

(A) 1/4 (B) 1/2
(C) 17/12 (D) 3/8

55. Compute 9 ÷ 3/4.

(A) 27/4 (B) 12
(C) 3/4 (D) 27

56. Half of a half of a half of a whole pizza is what fraction of the pizza?

(A) 1/2 (B) 1/6
(C) 3/2 (D) 1/8

57. A road 3/5 km long is being repaired; 2/3 of it is finished. How many kilometres are still unrepaired?

(A) 2/5 (B) 1/5
(C) 2/3 (D) 1/3

58. Which expression equals 4?

(A) 1/2 ÷ 1/8 (B) 1/2 × 4
(C) 8 ÷ 1/2 (D) 1/8 ÷ 1/2

59. A recipe for 4 people needs 2/3 cup of sugar. For 6 people you need (in cups):

(A) 1/2 (B) 4/9
(C) 1 (D) 8/9

60. Anil reads 1/5 of a book on Monday and 1/4 of the remaining pages on Tuesday. What fraction of the whole book has he read by Tuesday night?

(A) 9/20 (B) 2/5
(C) 1/5 (D) 3/5

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 8: Working with Fractions

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	D	31	A	41	A	51	C
2	D	12	A	22	B	32	A	42	D	52	C
3	C	13	B	23	D	33	D	43	D	53	A
4	A	14	B	24	B	34	C	44	D	54	B
5	C	15	C	25	C	35	A	45	A	55	B
6	C	16	C	26	D	36	D	46	C	56	D
7	D	17	B	27	D	37	D	47	B	57	B
8	A	18	D	28	A	38	B	48	B	58	A
9	C	19	A	29	B	39	C	49	A	59	C
10	B	20	A	30	C	40	D	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) "Product of 1/2 and 1/3" = 1/2 × 1/3 = (1×1)/(2×3) = 1/6. (Adding the two, 1/2 + 1/3 = 5/6, is the trap (A).)

2. (D) 3 × 1/4 = 1/4 + 1/4 + 1/4 = 3/4. Multiply the whole number into the numerator only; the denominator (the piece size) does not change. (3/12 wrongly multiplies the denominator too.)

3. (C) The word "of" means multiply, so "2/5 of 30" = 2/5 × 30. (Division and addition are the standard distractors.)

4. (A) The reciprocal swaps numerator and denominator: 3/7 → 7/3. (The reciprocal is not the negative.)

5. (C) 7 × 3/5 = 21/5. As a mixed number, 21 ÷ 5 = 4 remainder 1, so 4 1/5. (21/35 wrongly multiplies the denominator.)

6. (C) "2/3 of 3/5" = 2/3 × 3/5 = 6/15 = 2/5 in lowest terms. (Leaving it as 6/15-style unreduced or adding gives the distractors.)

7. (D) 5/6 ÷ 2/3 = 5/6 × 3/2 (flip the divisor) = 15/12 = 5/4. (5/9 comes from multiplying instead of flipping; 10/18 is the unreduced wrong product.)

8. (A) 3 ÷ 1/2 asks "how many halves fit in 3?" = 3 × 2 = 6. Dividing by a number below 1 gives a result larger than 3. (3/2 flips the wrong fraction.)

9. (C) Multiplying by a fraction LESS than 1 shrinks a number. Only 8 × 3/4 = 6 < 8; 8 × 5/4 = 10 and 8 × 9/8 = 9 are larger, and 8 × 1 = 8.

10. (B) A whole number 5 = 5/1, so its reciprocal is 1/5 (their product is 1). (5 itself is not its own reciprocal.)

11. (C) Dividing by 5/4 = multiplying by its reciprocal 4/5. So dividing by 5/4 matches multiplying by 4/5. (Dividing by 4/5 would equal multiplying by 5/4 — the reversed trap.)

12. (A) Area = length × breadth = 2/3 × 3/4 = 6/12 = 1/2 m^2. (5/7 and 6/7 come from adding numerators/denominators.)

13. (B) 2 1/2 = (2×2 + 1)/2 = 5/2. (4/2 forgets the +1; 21/2 mis-places digits.)

14. (B) Convert first: 2 1/2 = 5/2, 1 1/3 = 4/3. Product = 5/2 × 4/3 = 20/6 = 10/3 = 3 1/3. (Multiplying whole parts separately is the classic mixed-number error.)

15. (C) 4 1/2 ÷ 3/4 = 9/2 ÷ 3/4 = 9/2 × 4/3 = 36/6 = 6 shares. Dividing by a fraction below 1 yields more pieces than the acreage.

16. (C) Dividing by 1/4 means multiplying by 4 (its reciprocal) — TRUE. (B) 6 ÷ 1/2 = 12, not 3; (A) ×2/3 shrinks a number; (D) reciprocal of 1 is 1, not 0.

17. (B) "Half of a third" = 1/2 × 1/3 = 1/6 (of means multiply, not add). 1/2 + 1/3 = 5/6 is the trap (A).

18. (D) After the book, 1 − 1/3 = 2/3 remains. The pen costs 1/4 of 2/3 = 1/6. Left = 2/3 − 1/6 = 4/6 − 1/6 = 3/6 = 1/2. (5/12 = 1/4 + 1/6 confuses "spent" with "left".)

19. (A) Flour = 1 1/2 × 2 2/3 = 3/2 × 8/3 = 24/6 = 4 cups. (4 1/2 and 4 1/6 come from not converting both mixed numbers.)

20. (A) 3/4 × 20 = (3 × 20)/4 = 60/4 = 15. (60 forgets to divide by 4; 5 divides by 4 but drops the 3.)

21. (D) All share the multiplier 12, so the largest fraction wins: 5/6 > 3/4 > 2/3 > 1/2. Thus 12 × 5/6 = 10 is the largest.

22. (B) 4/9 × 3/8 = 12/72. Reduce by 12: 1/6. (12/72 is correct but not in lowest terms; 7/17 adds across.)

23. (D) Distance = $1/4 \times 3\ 1/2 = 1/4 \times 7/2 = 7/8$ km. (3 1/2 ignores the rate; 14 multiplies wrongly.)

24. (B) $1 \div 2/3 = 1 \times 3/2 = 3/2$. Flip the divisor and multiply; the answer exceeds 1 because you divide by less than 1.

25. (C) Glasses = $2/3 \times 30 = 20$. Both glasses and caps = $1/5 \times 20 = 4$. (20 stops at the first step; 6 = 1/5 of 30 skips the chaining.)

26. (D) $2/3 \div 4 = 2/3 \times 1/4 = 2/12 = 1/6$. Dividing by a whole number = multiplying by its reciprocal. (8/3 multiplies by 4 instead.)

27. (D) Red = $2/3 \times 6 = 4$ m, so 2 m remain. Blue = 1/2 of that rest = $1/2 \times 2 = 1$ m. (Taking 1/2 of the whole pole gives 3 — the trap.)

28. (A) Reciprocals multiply to 1: $2/5 \times 5/2 = 10/10 = 1$. So 2/5 and 5/2. (1/2 and 2/4 are equal, not reciprocal.)

29. (B) $3/4 \div 1/8 = 3/4 \times 8 = 24/4 = 6$ pieces. Many small 1/8 m pieces fit in 3/4 m. (3/32 multiplies instead of dividing.)

30. (C) 4/5 is less than 1, so $15 \times 4/5 = 12$ is LESS than 15. Multiplying by a proper fraction shrinks the number. (It is certainly more than 4/5, so (D) is false.)

31. (A) $(2/3) \times (9/10) \times (5/6) = (2\times9\times5)/(3\times10\times6) = 90/180 = 1/2$. (90/180 is correct but unreduced.)

32. (A) $2\ 1/4 \div 3/8 = 9/4 \div 3/8 = 9/4 \times 8/3 = 72/12 = 6$ glasses. (27/32 multiplies instead of dividing.)

33. (D) 3/4 of the number = 18, so number = $18 \div 3/4 = 18 \times 4/3 = 72/3 = 24$. Check: $3/4 \times 24 = 18$. ✓ (13 1/2 = 3/4 × 18 multiplies instead of dividing; 72 multiplies by 4 only.)

34. (C) Dividing by 1/3 = multiplying by its reciprocal 3, so $5 \div 1/3 = 5 \times 3 = 15$. (5 × 1/3 multiplies by the divisor itself.)

35. (A) $7/8 \times 8/7 = 56/56 = 1$ (a number times its reciprocal). (B) = 3/4; (C) = $1 \times 1/2 = 1/2$; (D) = $1 \times 1/2 = 1/2$.

36. (D) $1/2 \times 1/3 \times 1/4 = (1\times1\times1)/(2\times3\times4) = 1/24$. (3/24 = 3/24 adds numerators; 1/9 adds denominators-style.)

37. (D) Travelled 3/5 of 200 = 120 km, so remaining = $200 - 120 = 80$ km. (Equivalently $2/5 \times 200 = 80$.) (120 answers the wrong part.)

38. (B) $2\ 1/3 \div 1\ 1/6 = 7/3 \div 7/6 = 7/3 \times 6/7 = 42/21 = 2$. (49/18 multiplies the fractions; 14/18 mishandles the reciprocal.)

39. (C) A number times 1 is itself, so 3/8 × 1 = 3/8. The multiplicative identity is 1. (×0 gives 0; ×3/8 gives 9/64.)

40. (D) Whole wall ÷ rate = 1 ÷ 2/5 = 1 × 5/2 = 5/2 hours. Dividing by a rate below 1 takes more than 1 hour. (2/5 and 7/5 misuse the reciprocal.)

41. (A) After Riya, 1 − 1/4 = 3/4 remains. Sam ate 2/3 of that: 2/3 × 3/4 = 6/12 = 1/2 of the whole cake. (2/3 forgets it is "of the remainder".)

42. (D) 5/12 × 8 = (5 × 8)/12 = 40/12 = 10/3. (40/96 wrongly multiplies the denominator by 8 too.)

43. (D) FALSE statement: a fraction is NOT always larger than its reciprocal — 1/2's reciprocal is 2, which is bigger. (A), (B), (C) are all true: a proper fraction flips to >1, 0 has no reciprocal, and 1's reciprocal is 1.)

44. (D) Fraction filled = 3/4 × 1/3 = 3/12 = 1/4 in 1/3 hour. (9/4 divides instead of multiplying; 13/12 adds.)

45. (A) 1/2 + 1/3 = 5/6; then 5/6 × 6/5 = 30/30 = 1 (reciprocal pair). (Forgetting to add inside the bracket first gives the distractors.)

46. (C) Rose area = 3/10 of 5/6 = 3/10 × 5/6 = 15/60 = 1/4 acre. (8/16 and 15/16 mis-multiply or fail to reduce.)

47. (B) First number 9: 2/3 × 9 = 6. This equals 1/2 of the second, so 1/2 × x = 6 → x = 12. Check: 1/2 × 12 = 6. ✓ (6 stops at the first value; 27/4 mishandles the reciprocal.)

48. (B) Share each = 7/8 ÷ 7 = 7/8 × 1/7 = 7/56 = 1/8 m. (7/8 forgets to divide; 49/8 multiplies by 7.)

49. (A) Dividing by a number below 1 gives the biggest result: 6 ÷ 1/2 = 12, while 6 ÷ 1 = 6, 6 ÷ 2 = 3, 6 ÷ 3 = 2. So (A).

50. (A) 1 1/4 × 4/5 = 5/4 × 4/5 = 20/20 = 1. (5/4 and 4/5 forget to multiply the whole; 25/16 squares the wrong term.)

51. (C) In 2 days the worker paints 2 × 2/7 = 4/7, so unpainted = 1 − 4/7 = 3/7. (2/7 is one day's work; 4/7 is the painted part, not the leftover.)

52. (C) Area of a square = side² = (2/3)² = 2/3 × 2/3 = 4/9 cm². (8/3 and 4/3 mishandle the squaring; 4/6 fails to square the denominator.)

53. (A) We need 3/5 × ? = 1, so ? is the reciprocal 5/3 (since 3/5 × 5/3 = 1). (3/5 and 5 are tempting but only 5/3 works.)

54. (B) Poured out = 2/3 of 3/4 = 2/3 × 3/4 = 6/12 = 1/2 litre. (1/4 is the part LEFT; 3/8 mis-multiplies.)

55. (B) 9 ÷ 3/4 = 9 × 4/3 = 36/3 = 12. Dividing by a fraction below 1 gives more than 9. (27/4 multiplies by 3/4; 27 multiplies by 3 only.)

56. (D) Half of a half of a half = $(1/2)^3$ = 1/2 × 1/2 × 1/2 = 1/8 of the pizza. (1/6 adds the halves; 3/2 adds them as wholes.)

57. (B) Repaired = 2/3 of 3/5 = 2/3 × 3/5 = 6/15 = 2/5 km, so unrepaired = 3/5 − 2/5 = 1/5 km. (2/5 is the repaired length; 1/3 is the un-done fraction, not the length.)

58. (A) 1/2 ÷ 1/8 = 1/2 × 8 = 4. Checking the others: (B) 1/2 × 4 = 2; (C) 8 ÷ 1/2 = 8 × 2 = 16; (D) 1/8 ÷ 1/2 = 1/8 × 2 = 1/4. Only (A) equals 4. (The trap is thinking division must shrink the value — dividing by 1/8 multiplies by 8.)

59. (C) Sugar per person = 2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6 cup. For 6 people: 6 × 1/6 = 1 cup. (4/9 scales by 2/3 instead of unitary; 1/2 under-scales.)

60. (B) Monday: read 1/5, leaving 4/5. Tuesday: 1/4 of the remaining 4/5 = 1/4 × 4/5 = 4/20 = 1/5. Total read = 1/5 + 1/5 = 2/5. (9/20 mis-adds; 3/5 over-counts.)

Solutions complete: 60 / 60.